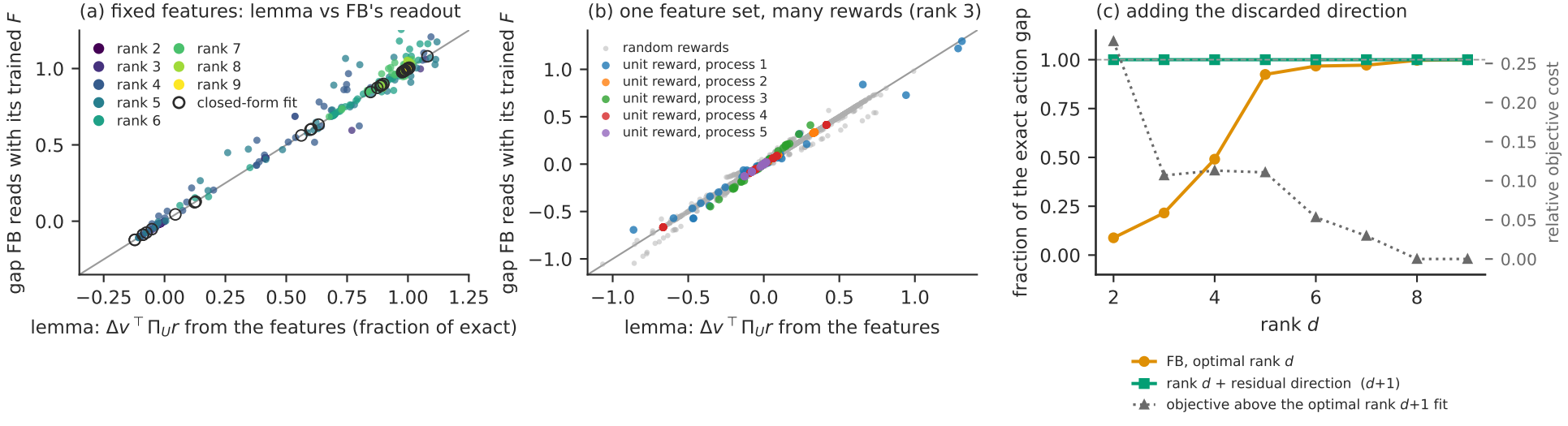

(a) fixed features: lemma vs FB's readout
gap FB reads with its trained F
lemma: Δv⊤ΠUr from the features (fraction of exact)
rank 2
rank 3
rank 4
rank 5
rank 6
rank 7
rank 8
rank 9
closed-form fit
(b) one feature set, many rewards (rank 3)
gap FB reads with its trained F
lemma: Δv⊤ΠUr from the features
random rewards
unit reward, process 1
unit reward, process 2
unit reward, process 3
unit reward, process 4
unit reward, process 5
(c) adding the discarded direction
fraction of the exact action gap
relative objective cost
rank d
FB, optimal rank d
rank d + residual direction (d+1)
objective above the optimal rank d+1 fit